Practice workbook · Draw a genetic diagram for a monohybrid or sex-linked cross

Full crosses for autosomal, codominant blood group, X-linked conditions and sex determination by X and Y chromosomes.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DRAW-MONOHYBRID-GENETIC-DIAGRAM

1. Question 5 · part c.i · 4 marks

November 2023 Paper 2

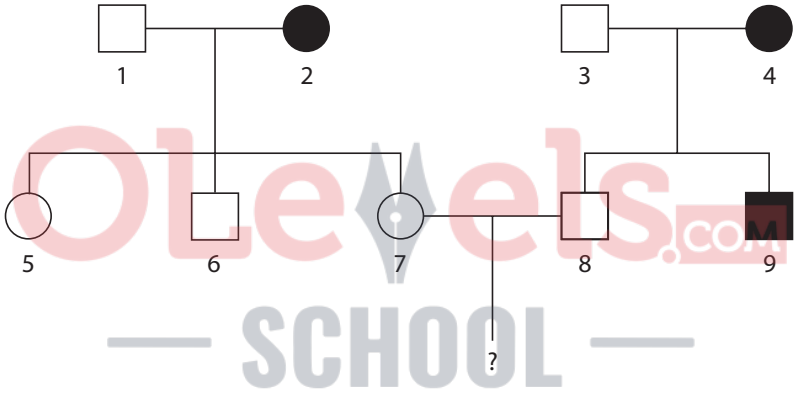
Worked-example retry · Attempt part c.i only.

5 Sickle cell anaemia is a condition caused by a recessive allele.

(a) Explain what is meant by the term **recessive allele**.

(2)

(b) The diagram shows a family pedigree. Some family members have sickle cell anaemia.



Key

- Male with sickle cell anaemia
- Male without sickle cell anaemia
- Female with sickle cell anaemia
- Female without sickle cell anaemia

(i) Give two people who are homozygous recessive.

(2)

(ii) Give the genotype of person 6.

Use the symbol **A** for the non-sickle cell allele and the symbol **a** for the sickle cell allele.

(1)

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1. Question 5 · part c.i · 4 marks

November 2023 Paper 2

Worked-example retry · Attempt part c.i only.

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(c) (i) People 7 and 8 in the pedigree produce a child.

Draw a genetic diagram to show the possible genotypes and phenotypes of this child.

(4)

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(ii) Calculate the probability that this child will be a boy with sickle cell anaemia.

(4)

probability = ...

(Total for Question 5 = 13 marks)

1. Question 5 · part c.i · 4 marks · Working and answer

Working and answer

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2. Question 9 · part c.i · 2 marks

June 2019 Paper 1

Further practice · Attempt part c.i only.

9 Inheritance of blood group involves codominant inheritance.

(a) (i) Which statement describes codominant inheritance in ABO blood groups?

(1)

- ☒ **A** the inheritance of two different alleles, both of which are expressed
- ☒ **B** the inheritance of two different alleles, only one of which is expressed
- ☒ **C** the inheritance of multiple alleles, only two of which are expressed
- ☒ **D** the inheritance of multiple alleles, only one of which is expressed

(ii) State the possible genotypes of a person with blood group A.

(1)

(b) (i) A person's blood group is determined by antigens.

These antigens are carbohydrate and protein molecules on the surface of red blood cells.

In 2007, a team of scientists used enzymes to convert blood groups A, B and AB into blood group O for transfusions.

Suggest how enzymes can convert blood groups A, B and AB into blood group O.

(3)

(ii) Suggest an advantage of producing blood group O using enzymes, compared with other methods of obtaining blood group O.

(1)

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P 5 8 5 6 8 A 0 2 4 2 8

2. Question 9 · part c.i · 2 marks

June 2019 Paper 1

Further practice · Attempt part c.i only.

(c) Haemophilia is a sex-linked blood disorder that reduces the ability of the blood to clot.

These are the genotypes of four offspring, P, Q, R and S.

P	Q	R	S
X^HX^h	X^hY	X^HX^H	X^HY

(i) Draw a genetic diagram to show how these offspring are produced from one set of parents.

(2)

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(ii) These parents are expecting another baby.

Determine the probability that this baby will have haemophilia.

(1)

probability = ...

(Total for Question 9 = 9 marks)

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2. Question 9 · part c.i · 2 marks · Working and answer

Working and answer

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3. Question 8 · part b.i · 4 marks

November 2020 Paper 2

Further practice · Attempt part b.i only.

8 Haemophilia is a sex-linked condition caused by a recessive allele X^h instead of the usual X^H.

Haemophilia is the result of a mutation.

(a) Explain what is meant by the term **mutation**.

(3)

(b) (i) A man without haemophilia and a woman who is a carrier of haemophilia have a child.

Draw a genetic diagram to show the possible genotypes and phenotypes of this child.

(4)

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3. Question 8 · part b.i · 4 marks

November 2020 Paper 2

Further practice · Attempt part b.i only.

(ii) The man and woman have a second child.

Calculate the probability that both children will have haemophilia.

(3)

(c) Explain why the man cannot be a carrier of haemophilia.

(5)

(Total for Question 8 = 15 marks)

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3. Question 8 · part b.i · 4 marks · Working and answer

Working and answer

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4. Question 8 · part b.i · 4 marks

June 2021 Paper 2

Further practice · Attempt part b.i only.

8 Skin and hair contain a pigment called melanin.

(a) (i) Describe the function of the pigment melanin in the skin.

(2)

(ii) State one other place, apart from hair and skin, where pigment is found in the body.

(1)

(b) A condition known as white forelock is caused by a dominant allele H.
People with this condition have a white patch in their hair.

The condition causes a deficiency of the skin pigment melanin.

A woman who is heterozygous for white forelock produces children with a father who does not have the condition.

(i) Draw a genetic diagram to show the possible offspring from this cross.

The condition is not sex-linked.

(4)

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4. Question 8 · part b.i · 4 marks

June 2021 Paper 2

Further practice · Attempt part b.i only.

(ii) Calculate the probability that these parents will produce a boy with the condition.

(3)

probability = ...

(c) White forelock is a result of a mutation in a molecule of DNA.

Describe how this mutation causes the white forelock phenotype.

(4)

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(Total for Question 8 = 14 marks)

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4. Question 8 · part b.i · 4 marks · Working and answer

Working and answer

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5. Question 7 · part b.ii · 6 marks

January 2022 Paper 2

Further practice · Attempt part b.ii only.

7 Huntington's chorea is a dominant genetic condition that results in the nervous system not working correctly. It is not a sex-linked condition.

Diagram 1 shows a family pedigree. Some members of the family have the condition.

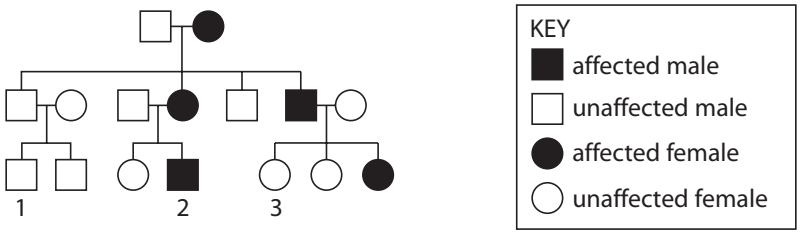


Diagram 1

(a) (i) Evaluate the evidence in the pedigree that supports the statement that this is not a sex-linked condition.

(2)

(ii) Evaluate the evidence in the pedigree to support the statement that this is a dominant condition.

(2)

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5. Question 7 · part b.ii · 6 marks

January 2022 Paper 2

Further practice · Attempt part b.ii only.

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(b) (i) A man has the same genotype as person 1 in the pedigree in Diagram 1.

A woman has the same genotype as person 3 in the pedigree in Diagram 1.

These two people are expecting a child.

Calculate the probability that the child would be a boy not showing the Huntington's condition, giving reasons for your answer.

(3)

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5. Question 7 · part b.ii · 6 marks

January 2022 Paper 2

Further practice · Attempt part b.ii only.

(ii) A man has the same genotype as person 2 in the pedigree in Diagram 1.

A woman has the same genotype as person 3 in the pedigree in Diagram 1.

Explain the possible phenotypes and genotypes, with respect to the Huntington's condition, of any offspring they produce.

(6)

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(Total for Question 7 = 13 marks)

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5. Question 7 · part b.ii · 6 marks · Working and answer

Working and answer

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6. Question 4 · part a.iii · 3 marks

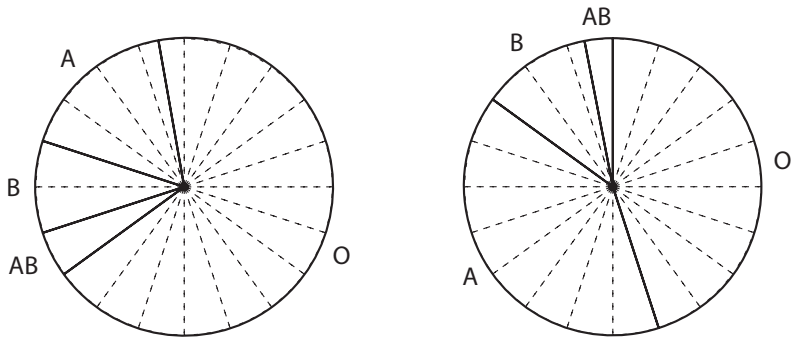
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Further practice · Attempt part a.iii only.

4 (a) Humans belong to one of four blood groups.

These blood groups are A, B, AB and O.

The pie charts show estimates of the percentages of people from Africa and the percentages of people from Europe in each blood group.



(i) Complete the tables to give the percentage of each blood group.

Some information has been given.

(2)

Africa

Blood group	Percentage of blood group
O	68
A	
B	
AB	5

Europe

Blood group	Percentage of blood group
O	45
A	
B	12
AB	

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P 7 0 9 7 3 A 0 1 2 2 4

6. Question 4 · part a.iii · 3 marks

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Further practice · Attempt part a.iii only.

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(ii) Some people need a blood transfusion.

Explain why people with blood group A cannot receive a blood transfusion from people with blood group B.

(3)

(iii) A male with the genotype I^AI^O and a female with the genotype I^BI^O have a child with the genotype I^OI^O.

Draw a genetic diagram to show how the child inherits the genotype I^OI^O.

(3)

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(iv) Determine the probability that a child from these parents will inherit the blood group AB.

(1)

probability = ...

6. Question 4 · part a.iii · 3 marks · Working and answer

Working and answer

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7. Question 8 · part b.iii · 2 marks

June 2022 Paper 2

Further practice · Attempt part b.iii only.

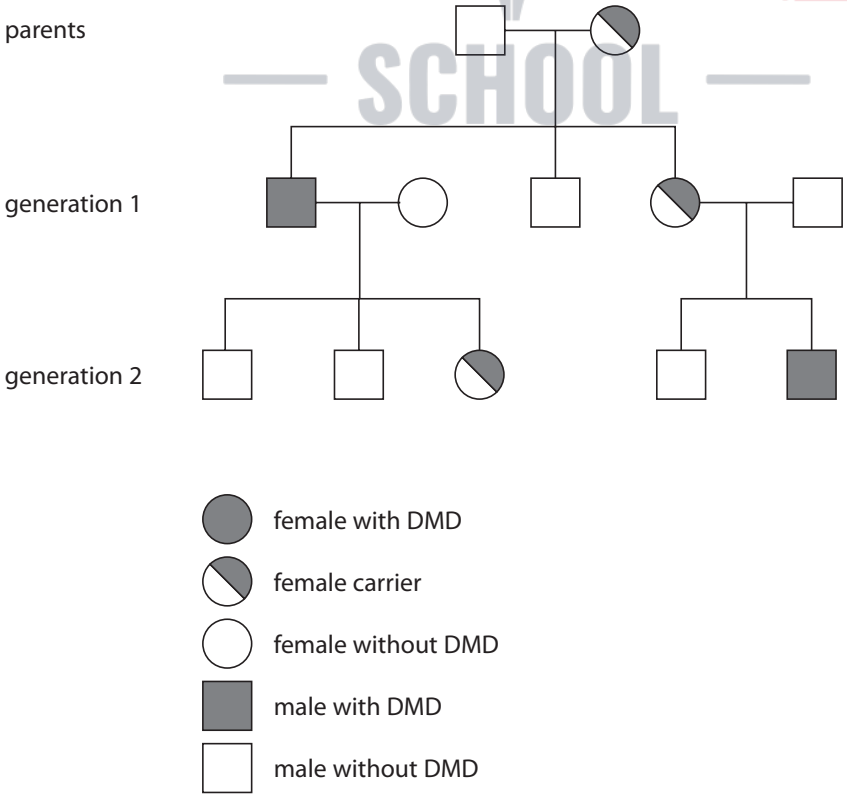
8 (a) Duchenne muscular dystrophy (DMD) is a sex-linked genetic disorder caused by a gene mutation.

The symptoms of DMD are caused by the absence of a protein called dystrophin.

Explain how a gene mutation can result in the absence of dystrophin.

(4)

(b) The diagram shows the inheritance of Duchenne muscular dystrophy in one family.



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P 7 0 9 7 4 A 0 1 6 2 0

7. Question 8 · part b.iii · 2 marks

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Further practice · Attempt part b.iii only.

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(i) Explain what is meant by a carrier of DMD.

(2)

(ii) Explain how the diagram shows that DMD is caused by a recessive allele.

(3)

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(iii) Draw a genetic diagram to show the inheritance of DMD from the parents to generation 1.

Use X^D for the dominant allele and X^d for the recessive allele.

(2)

(Total for Question 8 = 11 marks)

7. Question 8 · part b.iii · 2 marks · Working and answer

Working and answer

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8. Question 7 · part c.ii · 4 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

7 People with polycystic kidney disease have cysts on their kidneys.

Cysts are sacs of fluid and they affect the functioning of the kidney.

Polycystic kidney disease is caused by a mutation of a gene.

(a) Explain how a gene mutation can affect the phenotype of an individual.

(3)

(b) Polycystic kidney disease can cause kidney failure.

Kidney failure can be treated using dialysis or a transplant.

(i) Describe the benefits of a transplant rather than dialysis as a treatment for kidney failure.

(2)

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8. Question 7 · part c.ii · 4 marks

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Further practice · Attempt part c.ii only.

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(ii) In one year in the United Kingdom there were 4733 organ donations.

Of these donations, 3233 were kidney donations.

Calculate the percentage of organ donations that are kidney donations.

(2)

percentage = ... %

(iii) The number of kidney donations is much higher than the number of other organ donations such as heart or liver.

Give two reasons for the higher number of kidney donations.

(2)

1 .....

.....

2 .....

.....

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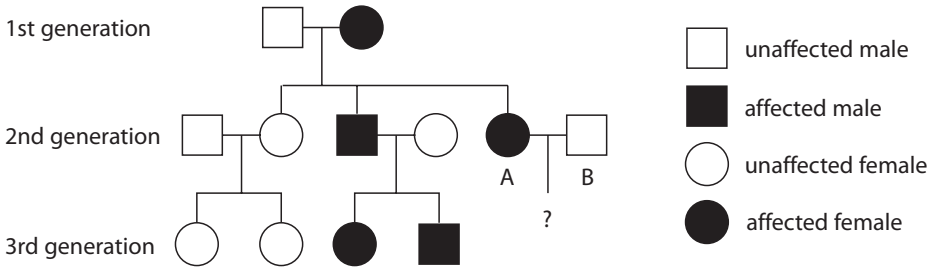
8. Question 7 · part c.ii · 4 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

(c) The diagram shows a family pedigree. Some of the family have polycystic kidney disease.

This disease is caused by a dominant allele that is not sex-linked.



(i) Determine the ratio of the females shown in the 3rd generation that are affected with polycystic kidney disease.

(1)

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ratio = ...

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8. Question 7 · part c.ii · 4 marks

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Further practice · Attempt part c.ii only.

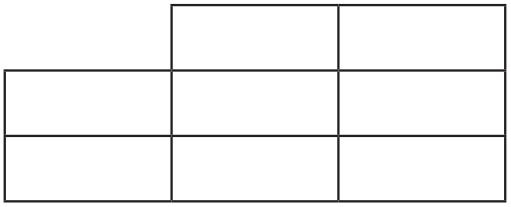
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(ii) Explain the possible genotypes and phenotypes for the offspring of individual A and individual B.

You should complete the genetic diagram as part of your answer.

(4)



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(Total for Question 7 = 14 marks)

8. Question 7 · part c.ii · 4 marks · Working and answer

Working and answer

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9. Question 3 · part b · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b only.

3 (a) Polycystic kidney disease (PKD) is an inherited condition caused by a gene mutation.

PKD affects the function of the kidneys.

State what is meant by a gene mutation.

(1)

.....

.....

(b) PKD occurs when a child inherits a faulty allele from one parent.

One parent has a genotype Hh and the other parent has a genotype hh.

The parents have four children.

Draw a genetic diagram to show the possible genotypes of the four children.

(2)

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(c) Some people with PKD need a kidney transplant.

Give one advantage and one disadvantage of a kidney transplant.

(2)

Advantage

.....

.....

Disadvantage

.....

.....

(Total for Question 3 = 5 marks)

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9. Question 3 · part b · 2 marks · Working and answer

Working and answer

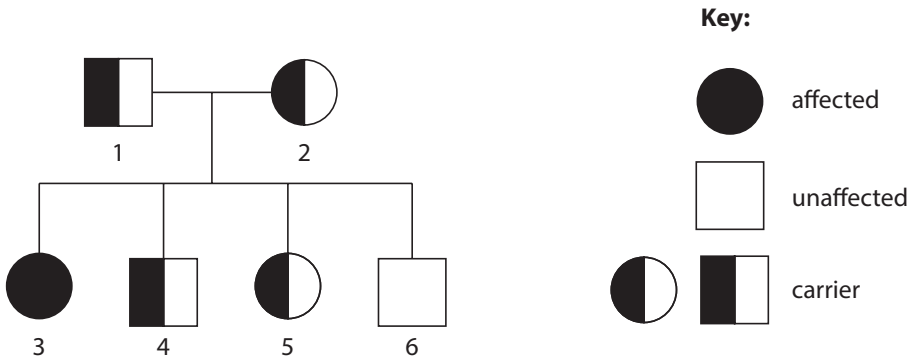
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10. Question 8 · part a.iv · 4 marks

June 2024 Paper 2

Further practice · Attempt part a.iv only.

8 (a) The diagram shows the inheritance of cystic fibrosis in a family.



(Source: Adapted from https://cysticfibrosiszaw2.weebly.com/)

(i) Which individual is homozygous recessive?

(1)

- ☒ **A** 1
- ☒ **B** 3
- ☒ **C** 5
- ☒ **D** 6

(ii) How many individuals have a heterozygous genotype?

(1)

- ☒ **A** 1
- ☒ **B** 2
- ☒ **C** 3
- ☒ **D** 4

(iii) Explain why carriers of cystic fibrosis do not have symptoms of the disease.

(2)

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10. Question 8 · part a.iv · 4 marks

June 2024 Paper 2

Further practice · Attempt part a.iv only.

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(iv) Draw a genetic diagram to show how two parents can have a 1 in 4 chance of having a child with cystic fibrosis.

Use F for the non-cystic fibrosis allele and f for the cystic fibrosis allele.

(4)

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(b) Describe how gene therapy can be used to treat cystic fibrosis.

(3)

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(Total for Question 8 = 11 marks)

10. Question 8 · part a.iv · 4 marks · Working and answer

Working and answer

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11. Question 6 · part b · 4 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

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6 (a) Red-green colour blindness is caused by a genetic mutation.

The table shows the percentage of red-green colour blind males and colour blind females in different regions.

Equal numbers of males and females were tested in each region.

Region	Total number of people tested	Percentage of colour blind males	Percentage of colour blind females
A	150	10.00	1.00
B	16 000	6.60	0.40
C	5000	0.80	0.01

(i) Calculate the number of colour blind males in the test group in region C.

(3)

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number of males = ...

(ii) State why the results from region A are not considered to be reliable.

(1)

.....

P 7 4 6 2 8 A 0 1 5 2 0

11. Question 6 · part b · 4 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

(iii) Justify the statement that colour blindness is a recessive sex-linked condition.

(5)

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11. Question 6 · part b · 4 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

(b) A woman who is heterozygous for the red-green colour blind condition has a child. The father does not have the condition.

Use a genetic diagram to determine the possible genotypes and phenotypes of the child.

Use B to represent the allele for normal vision and b to represent the allele for the colour blind condition.

(4)

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(Total for Question 6 = 13 marks)

11. Question 6 · part b · 4 marks · Working and answer

Working and answer

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12. Question 2 · part b.i · 3 marks

January 2023 Paper 1

Further practice · Attempt part b.i only.

2 (a) A human sperm cell has a tail, a mid-piece containing organelles and a head with a nucleus.

(i) Give the function of the tail.

(1)

(ii) Which of these, found in the mid-piece, release energy?

(1)

- ☒ **A** chromosomes
- ☒ **B** endoplasmic reticulum
- ☒ **C** mitochondria
- ☒ **D** ribosomes

(iii) The process of meiosis is involved in the production of sperm.

Explain the importance of meiosis in sperm production.

(3)

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12. Question 2 · part b.i · 3 marks

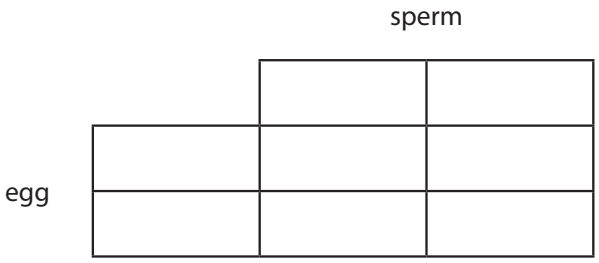
January 2023 Paper 1

Further practice · Attempt part b.i only.

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(b) (i) Complete the genetic diagram to show how the sex of a person is determined by the chromosomes in the nucleus of the egg and the nucleus of the sperm.

(3)



(ii) The sperm head has to enter the egg cell so the nuclei can fuse.

Describe how the sperm head enters the egg cell.

(2)

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(Total for Question 2 = 10 marks)

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12. Question 2 · part b.i · 3 marks · Working and answer

Working and answer

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